

Question 1 — Science

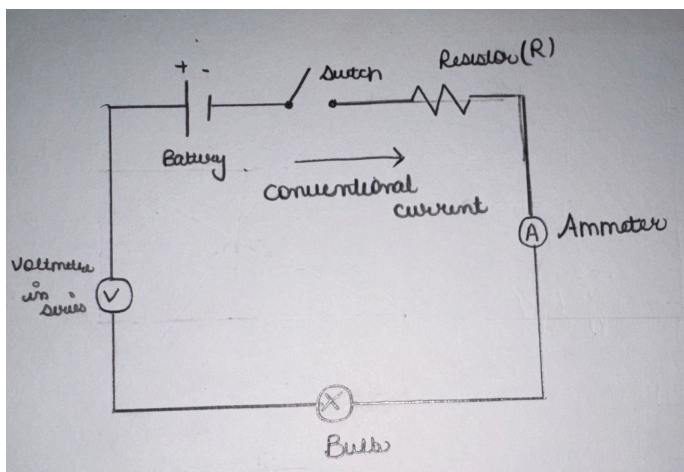
An electric circuit is a path made of wires which joins all the parts together in one loop. The parts I have used are a battery, a switch, a bulb, a resistor, an ammeter and a voltmeter, and they are all joined one after the other in the same loop.

The switch is the part which stores the energy and releases it slowly into the wires when it is pressed. The battery's job is to control how fast the current moves. The resistor is used to make the current stronger before it reaches the bulb, which is why the bulb needs a resistor next to it.

The voltmeter is connected in series in the loop, and the ammeter is also connected in series, because both instruments have to have the current passing through them in order to work.

The current comes out from both terminals of the battery at the same time and meets at the bulb, which is what makes the bulb light up.

If we increase the resistance then the current becomes stronger, because the resistor gives the current an extra push. So a circuit with a bigger resistor will always have a brighter bulb than a circuit with a smaller one.



Question 2 — English

Technology is bad for students and it should not be used for studying.

Students today use phones and computers all the time and this is the reason why results are getting worse everywhere. Everyone knows that reading from a book is better than reading from a screen. Teachers also say the same thing. If technology was actually good for learning then students would be much cleverer now than they were before, and they are not.

Therefore technology is bad and students should stay away from it.

Question 3 — Economics

(a) The graph is drawn below. Price is on the vertical axis and quantity is on the horizontal axis. The demand curve goes down and the supply curve goes up.

(b) The equilibrium of the market is at ₹40 and 40 units, which is the point where the two lines cross on my graph.

(c) When the price is below the equilibrium there is a surplus in the market, because the producers are supplying more than the consumers want. When the price is above the equilibrium there is a shortage, because the consumers want more than the producers are supplying.

(d) If the cost of production increases, the producers will need to earn more money, so they will produce a larger quantity to cover the extra cost. The supply curve therefore shifts to the right.

The new equilibrium will be at a lower price and a higher quantity than before.

